Grade 9 Technology

Exam Study Notes · November 2026
Based on the school demarcation (Afbakening)

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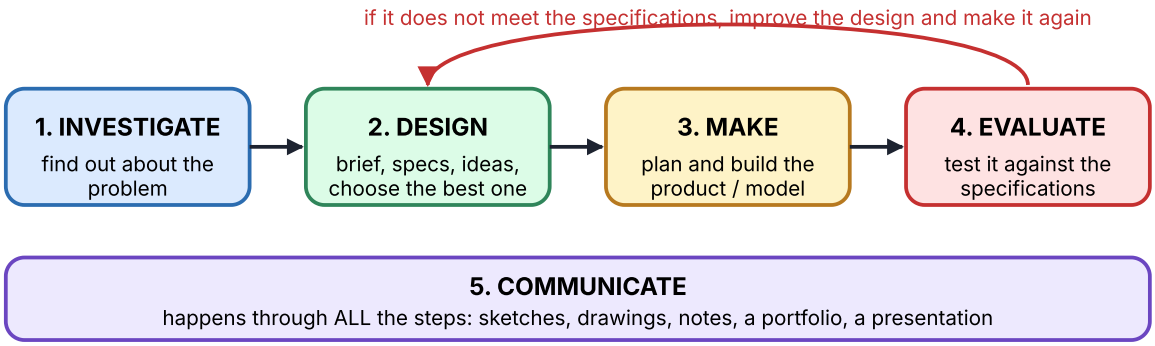
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1. The Technological Process

Technology is using knowledge, skills and resources to **solve problems** and meet people's **needs and wants**. The **technological process** (also called the **design process**) is the set of steps you follow to get from a problem to a working solution. In South Africa it is remembered as **IDMEC**.



The technological (design) process – IDMEC

Step 1: Investigate

You first have to **understand the problem** properly before you try to solve it.

- **The scenario / problem statement:** a short story that describes a situation and the problem in it. E.g. "Learners at a rural school walk home in the dark and cannot see the path."
- **Research:** find out more – look at **existing products** that solve similar problems (what is good and bad about them), read, use the internet, ask people (interviews, questionnaires).
- **Ask the right questions:** Who is it for (the **user**)? Where will it be used? What must it do? What materials and tools are available? How much may it cost? How much time do I have?

Step 2: Design

The design brief

A **design brief** is a short, clear statement that says **what you are going to design and make, for whom, and why**. It does not describe exactly what the product looks like.

> **Example design brief:** "I will design and make a battery-powered night light for learners who walk home in the dark, so that they can see the path safely."

Specifications and constraints

	Meaning	Example (night light)
Specifications	The **requirements** the product must meet – what it must do, look like and be made of. You use them later to test (evaluate) the product.	It must switch on automatically when it gets dark. It must be light enough to carry. It must be waterproof.
Constraints	The **limits** on the design – things you can't change and must work within	It must cost less than R100. It must be finished in 3 weeks. Only a 9 V battery may be used.

Specifications are often grouped under headings such as **function** (what it must do), **aesthetics** (how it looks: colour, shape, texture), **ergonomics** (comfortable and easy for people to use), **safety**, **materials** and **cost**.

Ideas and the final design

1. Draw **at least 2–3 different possible solutions** as quick, labelled **freehand sketches**.
2. Compare each idea against the specifications and constraints (often in a table) and choose the **best idea**, giving **reasons**.
3. Draw the final design neatly – often as an **isometric** drawing (what it looks like) and **orthographic** drawings (exact sizes) – see chapters 2 and 3.

Step 3: Make

- Make a **planning** chart: list the steps in the correct order, the **tools** and **materials** for each step, and how long each will take.
- Draw up a **list of materials** (with sizes and quantities) and a **cost** list.
- Build the product or a **model** (a smaller or simpler version) using the correct tools **safely**: wear safety glasses, use tools only for their purpose, keep the workspace neat, and switch off electrical equipment.

Step 4: Evaluate

To **evaluate** means to **test and judge** the product. Go back to your **specifications** and check each one: does the product meet it? Ask questions such as: Does it work? Is it safe? Does it look good? Did it stay within the cost and time? What would I improve? If it does not meet a specification, you **change the design** and improve it.

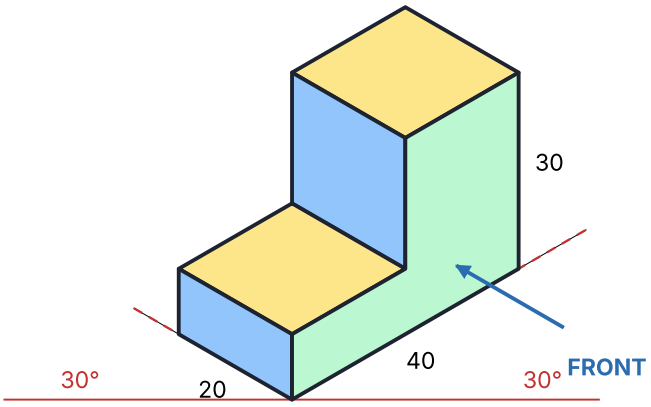
Step 5: Communicate

You record and share your work **during every step** – not just at the end – using sketches, drawings, notes, tables, photos, a **portfolio** and/or a presentation. This shows your thinking and lets someone else understand or rebuild your product.

> **Remember:** the process is **not a straight line**. You often go back to an earlier step (e.g. evaluate → redesign), until the product meets the specifications.

2. Isometric Drawings

An **isometric drawing** is a **3D (pictorial) drawing** that shows an object the way you see it – with three faces visible (front, side and top) at once. "Iso-metric" means "equal measure": the three axes are equally foreshortened, so you can measure **true lengths** along them.



How to read it

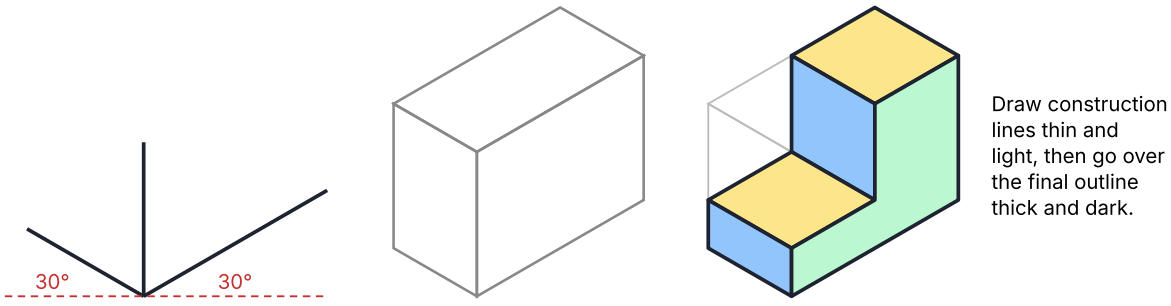
- vertical edges stay vertical
- width and depth edges are drawn at 30°
- lengths along the 3 axes are measured true size
- no hidden (dashed) lines
- 3 faces seen: front, side and top

An isometric drawing of an L-shaped block (sizes in mm). The same block is drawn orthographically in chapter 3.

Rules of isometric drawing

Rule	What it means
Vertical lines stay vertical	All the height edges are drawn straight up and down.
Horizontal lines are drawn at 30°	The width and depth edges are drawn at **30° to the horizontal** – one set going up to the right, the other up to the left. Use a **30°/60° set square** on a T-square or ruler.
Measure true lengths along the axes	A 40 mm edge is drawn 40 mm long (or to scale) – but **only along the three isometric axes**. Sloping (non-isometric) lines can't be measured directly; find their two end points and join them.
Parallel edges stay parallel	Edges that are parallel on the real object are parallel on the drawing.
No hidden lines	You only draw the edges you can see. Hidden edges are left out.

How to draw an isometric drawing (the box / crating method)



1. Draw the 3 axes
2. Draw a light box (crate)
3. Mark sizes, cut away, darken

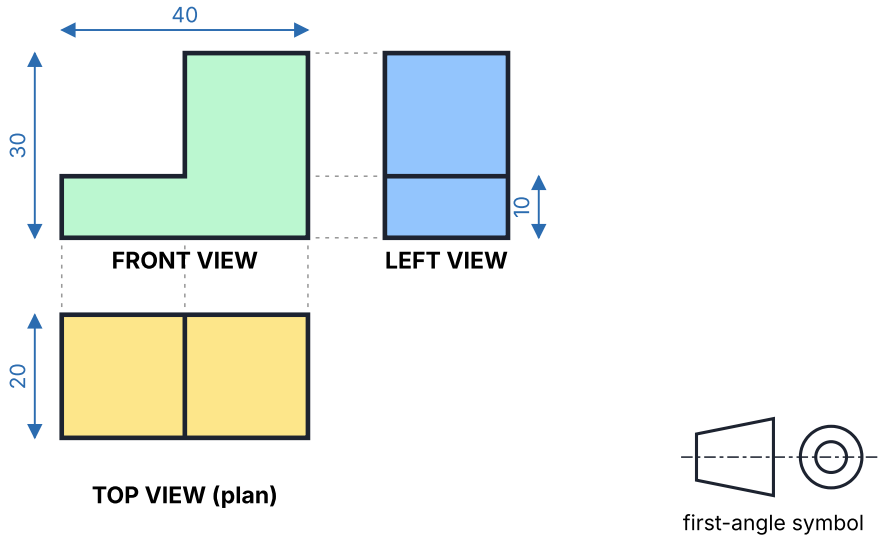
Drawing an isometric view in three steps

1. Draw the **three isometric axes** from one corner point: one vertical line and two lines at 30° (left and right).
2. Using the overall **width, depth and height**, draw a light **box** (crate) that the whole object fits into.
3. Measure and mark the sizes of the details along the axes, **cut away** the parts that are not there (e.g. the step of the L-shape), then **darken the outline**. Rub out or leave the light construction lines.

> **Isometric grid paper** has dots or lines already at 30° and vertical – you just follow the lines and count the squares. A circle on an isometric drawing looks like an **ellipse** (a flattened circle).

3. Orthographic Drawings

An **orthographic drawing** (orthographic projection) shows an object as a set of **2D views**, each looking **straight at one side**. It is a **working drawing**: it gives the exact **shape and sizes** needed to make the object. In South Africa we use **first-angle projection**.



First-angle orthographic projection of the L-shaped block from chapter 2 (sizes in mm)

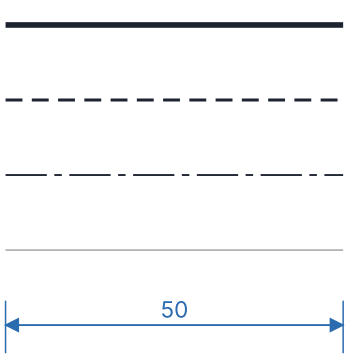
The three views and where they go (first angle)

View	You look at the object …	Placed …
Front view (elevation)	straight from the front – it usually shows the most detail / the main shape	top left – all other views are placed in relation to it
Top view (plan)	straight down from above	**below** the front view
Left view (end view)	from the left side	to the **right** of the front view

> **First angle = "the view is placed on the opposite side to where you look from".** You look from the top → draw it underneath. You look from the left → draw it on the right.

- The views **line up** with each other: the front and top views have the **same width**; the front and left views have the **same height**; the top and left views both show the **depth**. Thin projection lines help you line them up.
- Each view shows only **two** of the three measurements (e.g. the front view shows width and height, not depth).
- Write the **first-angle symbol** (a cone seen from the side and the end) on the drawing so the reader knows which method was used.

Line types



Outline – thick, dark, continuous: the visible edges of the object

Hidden line – thin, short dashes: edges you can't see from that side

Centre line – thin long-short chain: centre of holes, circles and symme

Construction / projection line – very thin and light

Dimension line – thin, with arrowheads touching the thin extension (projection) lines; size written above it

The main line types used in technical drawings

Dimensioning rules

- All sizes are in **millimetres (mm)**; do not write "mm" after every number – write "All dimensions in mm" once.
- The dimension line is thin with **arrowheads at both ends** that touch the **extension lines**; extension lines start with a small gap away from the object.
- Write the number **above the middle** of the dimension line, readable from the **bottom** or the **right-hand side** of the page.
- Place dimensions **outside** the views where possible, and do not repeat the same dimension on two views.
- A **scale** must be stated: 1:1 = full size; 1:2 = half the real size (reduced); 2:1 = twice the real size (enlarged).

Comparing isometric and orthographic drawings

	Isometric drawing	Orthographic drawing
Type	3D, pictorial – one picture	2D – several separate views
Shows	What the object looks like as a whole	The exact shape and sizes of each side
Angles	Vertical lines + lines at 30°	Only horizontal and vertical lines (90°)
Hidden detail	Not shown	Shown with dashed hidden lines
Used for	Presenting the design to a client	Making (manufacturing) the product

4. Electrical Components + Control

Basic ideas

An **electric circuit** is a **closed path** along which electric current can flow. Every working circuit needs:

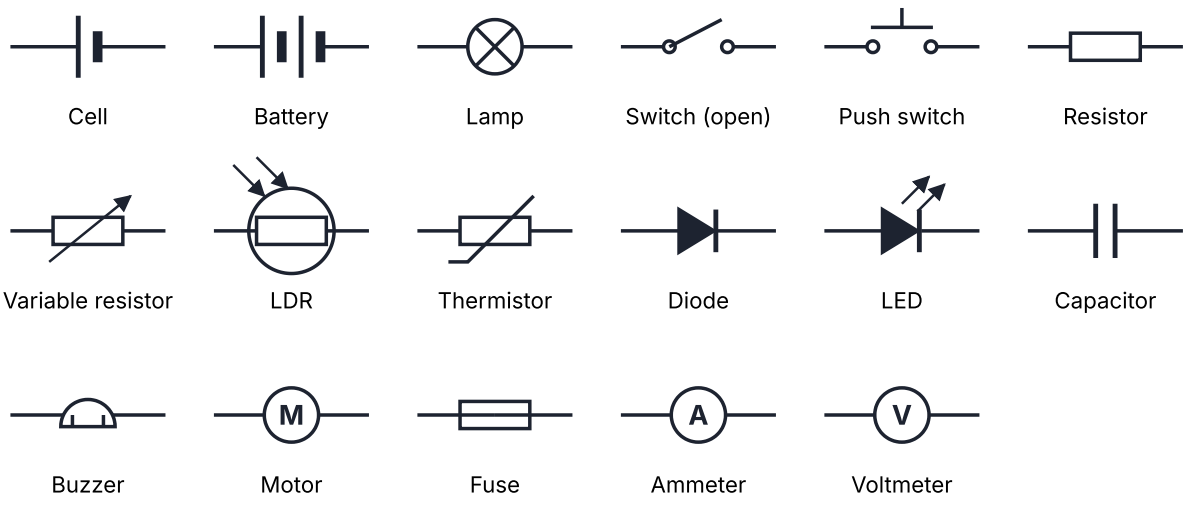
1. an **energy source** (cell or battery) – pushes the current,
2. **conductors** (wires) – the path for the current,
3. a **load / output device** (lamp, LED, buzzer, motor) – changes electrical energy into another form (light, sound, movement),
4. usually a **switch** – the **control** that opens or closes the circuit.

Quantity	What it is	Unit	Measured with
Current (I)	The flow of charge (electrons) through the circuit	ampere (A)	**Ammeter** – connected in **series**
Voltage / potential difference (V)	The "push" (energy) that drives the current	volt (V)	**Voltmeter** – connected in **parallel** across the component
Resistance (R)	How much a component **opposes** (resists) the current	ohm (Ω)	Ohmmeter / multimeter

Conventional current is shown flowing from the **positive (+)** terminal of the source, around the circuit, to the **negative (–)** terminal. (Electrons actually move the other way.)

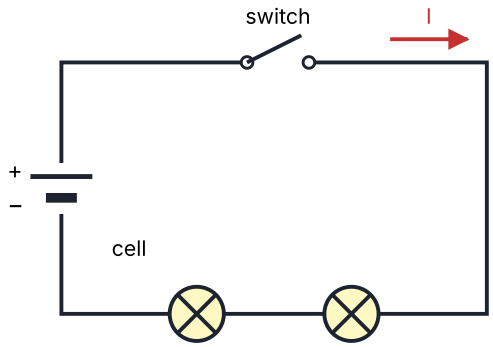
Circuit symbols

Circuit diagrams use standard **symbols** so that anyone can read them. Wires are drawn as straight lines with square corners; a **dot** shows where wires are joined.

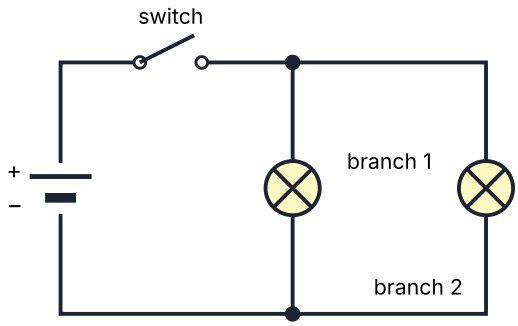


Standard circuit symbols (the longer, thin plate of a cell is the + terminal)

Series and parallel circuits



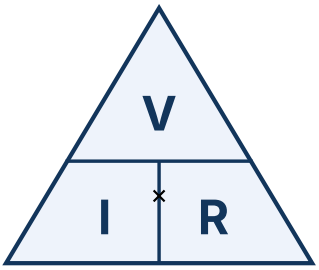
Series circuit – one path



Parallel circuit – more than one path

	Series circuit	**Parallel circuit**
Paths	Only **one** path for the current	**Two or more** paths (branches)
Current	The **same** everywhere in the circuit	Splits up between the branches and adds up again
Voltage	Is **shared** between the components	Each branch gets the **full** voltage of the source
Brightness	More lamps → each lamp is **dimmer**	Lamps stay **bright** (but the battery runs flat faster)
If one lamp breaks	The circuit is broken – **all** lamps go off	The other lamps **stay on** (used in house wiring)
Cells	Cells in series: voltages **add up** (2 × 1,5 V = 3 V)	Cells in parallel: same voltage, but last longer

Ohm's law



Cover the one you want

Ohm's law: the current through a resistor is directly proportional to the voltage across it (at a constant temperature).

V = I × R I = V ÷ R R = V ÷ I
V = voltage in volts (V), I = current in amperes (A), R = resistance in ohms (Ω)

Example: A 9 V battery is connected to a 450 Ω resistor. What current flows?
I = V ÷ R = 9 ÷ 450 = **0,02 A** (= 20 mA)
Example: 0,5 A flows through a lamp when 6 V is across it. What is its resistance?
R = V ÷ I = 6 ÷ 0,5 = **12 Ω**

More resistance → less current (for the same voltage). 1 kΩ = 1 000 Ω; 1 mA = 0,001 A.

Electronic components

Resistors

A **resistor limits (reduces) the current** in a circuit, e.g. to protect an LED from burning out. A **fixed resistor** has one value; a **variable resistor** (potentiometer) can be adjusted by turning a knob – e.g. a volume control or a dimmer.



1 0 × 100 = 1 000 Ω = 1 kΩ (± 5%)

Reading the colour code
Band 1 = first digit
Band 2 = second digit
Band 3 = multiplier (number of zeros)
Band 4 = tolerance (gold ± 5%, silver ± 10%) – read from the other end

Resistor colour code

Colour	Black	Brown	Red	Orange	Yellow	Green	Blue	Violet	Grey	White
Number	0	1	2	3	4	5	6	7	8	9

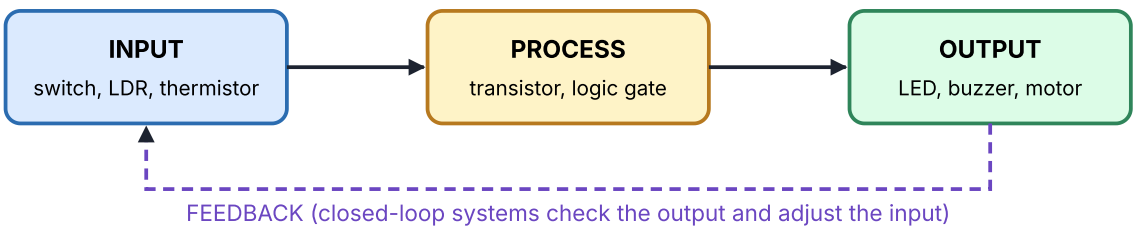
More examples:
yellow – violet – red = 4 7 × 100 = **4 700 Ω = 4,7 kΩ**
red – red – brown = 2 2 × 10 = **220 Ω**
brown – black – orange = 1 0 × 1 000 = **10 000 Ω = 10 kΩ**

Other components

Component	What it does / how it works	Used in
LDR (light-dependent resistor)	A resistor whose resistance **changes with light**. **Bright light → low resistance**; **dark → high resistance**. It is an **input sensor**.	Street lights and night lights that switch on automatically at night
Thermistor	A resistor whose resistance **changes with temperature**. (Most common type: **hotter → lower resistance**.) Input sensor.	Fire alarms, thermostats, fridge and car temperature sensors
Diode	Lets current flow in **one direction only** (like a one-way valve). The arrow in the symbol points the way current can flow; the bar is the cathode (−).	Protecting circuits if a battery is put in the wrong way; changing AC to DC
LED (light-emitting diode)	A diode that **gives off light** when current flows through it in the correct direction. The longer leg is the anode (+). It needs a **series resistor** to limit the current, otherwise it burns out. Uses very little energy and lasts long.	Indicator lights, torches, traffic lights, screens
Capacitor	**Stores electric charge** for a short time and then releases it. Value in farads (F), usually µF. Can be used to make a **time delay**.	Camera flash, timers, smoothing power supplies
Transistor	An electronic **switch** (and amplifier) with three legs: **base (b)**, **collector (c)** and **emitter (e)**. A **small current** into the base switches on a **much bigger current** from collector to emitter. No moving parts.	Automatic switching circuits, amplifiers, computers
Buzzer, **motor**, **lamp**	**Output devices**: change electrical energy into sound, movement and light.	Alarms, fans, lights
Fuse	A thin wire that **melts and breaks the circuit** if too much current flows – a **safety** device.	Plugs, cars, distribution boards

Control

Control means deciding **when** and **how** an output works. Every controlled system can be shown as a **systems diagram**.



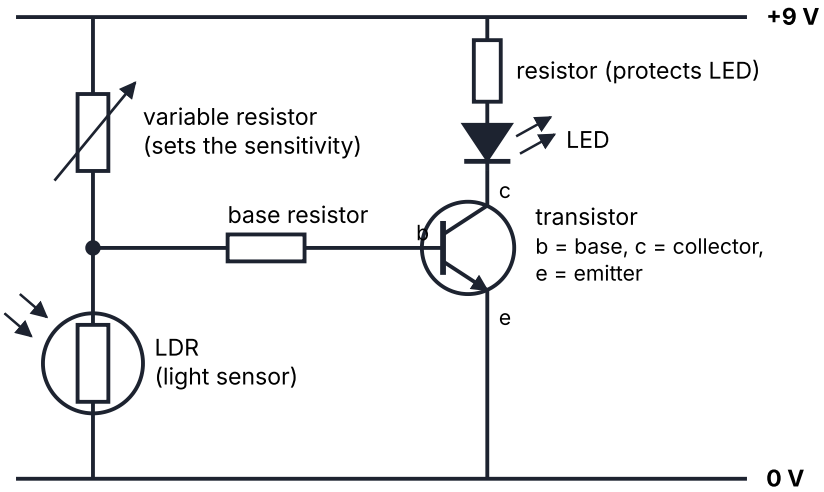
Example – night light: it gets dark (LDR) → transistor switches on → LED lights up

A systems diagram: input → process → output

Switches as control devices

Switch	How it works	Example
Toggle / rocker switch (SPST)	Stays on or off until you change it. SPST = single pole, single throw: one circuit, on/off.	Light switch
Push-to-make switch	Only closes (on) **while you push** it; springs back off.	Doorbell, keyboard keys
Push-to-break switch	Normally on; **breaks** the circuit while you push it.	Fridge door light (door pushes it off)
Micro switch	A small switch with a lever that needs very little force to operate.	Microwave door, vending machines
Reed switch	Two thin metal strips in a glass tube that touch when a **magnet** comes close.	Burglar alarms on doors and windows
Tilt switch	A ball or mercury closes the contacts when it is **tilted**.	Car alarms, toys

A transistor switching circuit (automatic night light)

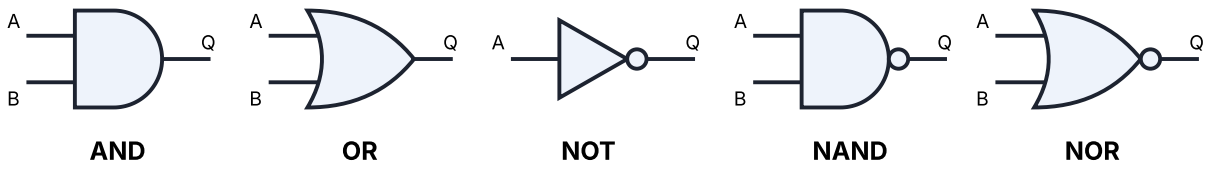


A transistor switch controlled by an LDR: the LED comes on when it gets dark

1. The variable resistor and the LDR form a **voltage divider**: the 9 V is shared between them according to their resistances.
2. **In the light**, the LDR's resistance is **low**, so it gets only a small share of the voltage. The voltage at the base is too low – the transistor stays **off** and the LED is off.
3. **In the dark**, the LDR's resistance becomes **high**, so it gets a big share of the voltage. The base voltage rises (above about 0,7 V), a small base current flows, the transistor switches **on**, and a bigger current flows from collector to emitter through the LED – the **LED lights up**.
4. Swap the LDR for a **thermistor** and the same circuit becomes a **temperature alarm** (with a buzzer as output).

Logic gates

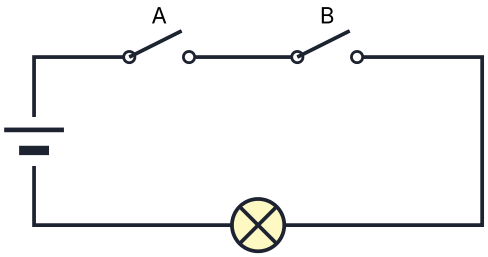
Logic gates are control devices that make **decisions**. Each input and output is either **1 (on, high)** or **0 (off, low)**. A **truth table** shows the output for every combination of inputs.



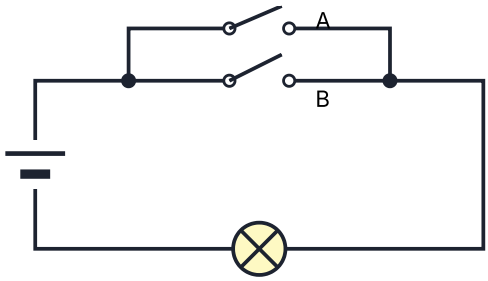
Logic gate symbols (A, B = inputs; Q = output; the small circle means "NOT")

A	B	AND	OR	NAND	NOR		A	NOT
0	0	0	0	1	1		0	1
0	1	0	1	1	0		1	0
1	0	0	1	1	0			
1	1	1	1	0	0			

Gate	Rule	Switch circuit that does the same	Everyday example
AND	Output is 1 only if **A AND B** are both 1	Two switches in **series**	A machine that only runs if the guard is closed AND the start button is pressed
OR	Output is 1 if **A OR B (or both)** is 1	Two switches in **parallel**	A bus bell that rings if any of the buttons is pushed
NOT (inverter)	Output is the **opposite** of the input	A push-to-break switch	A light that comes on when it is NOT light outside
NAND	NOT-AND: the **opposite of AND** – 0 only when both inputs are 1		
NOR	NOT-OR: the **opposite of OR** – 1 only when both inputs are 0		



AND: switches in series – both must be closed



OR: switches in parallel – either one works

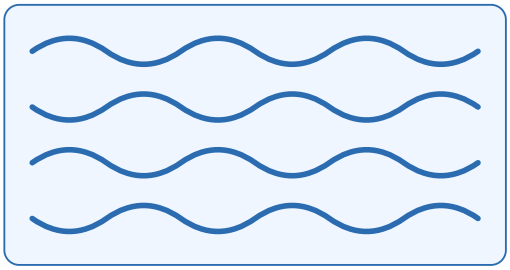
5. Plastics

Plastics are **synthetic (man-made) materials** made mostly from **crude oil** (and natural gas). They are made of very long chain-like molecules called **polymers**, built from many small repeating units called **monomers** joined together (e.g. many ethene monomers → polyethene). Plastics can be shaped (moulded) when soft, which is where the name comes from (*plastikos* = able to be moulded).

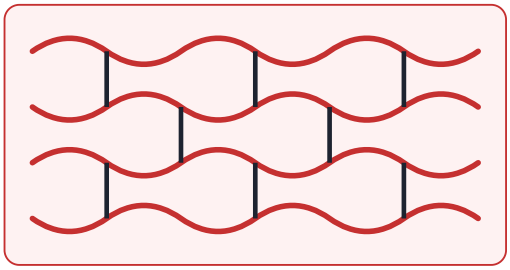
Properties of plastics

Property	Why it is useful
Lightweight (low density)	Cheap to transport; good for bottles, car parts
Good insulator of electricity and heat	Cable covers, plugs, switches, pot handles
Waterproof and resistant to rust / corrosion and many chemicals	Pipes, containers, outdoor furniture
Easy to shape (mould) into complex shapes; can be any colour	Mass-produced toys, casings, bottles
Durable – strong for its weight, tough, does not rot	Long-lasting products ... but also why plastic **pollution** lasts hundreds of years
Cheap to produce in large quantities	Packaging, disposable items

Thermoplastics and thermosetting plastics



Thermoplastic: *separate chains with weak forces between them – they slide apart when heated*

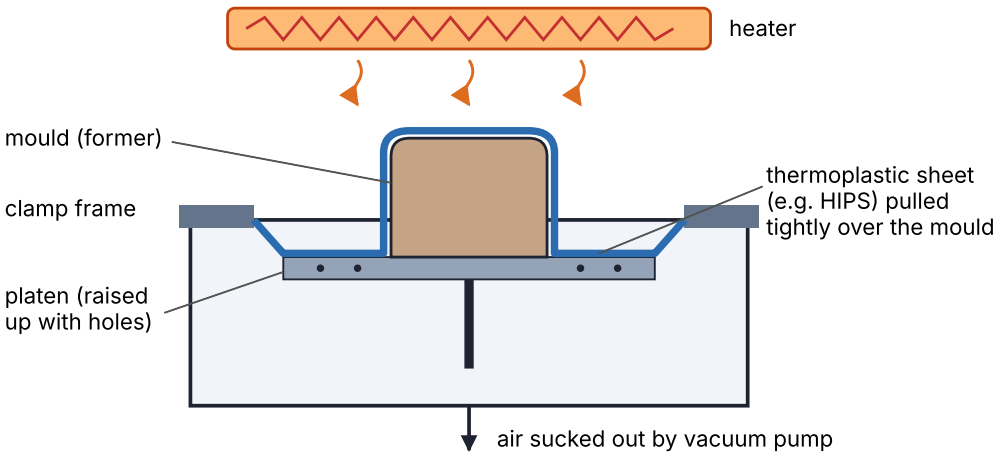


Thermosetting plastic: *chains joined by strong* ***cross-links*** *– a rigid network that can't soften*

	Thermoplastics	Thermosetting plastics (thermosets)
When heated	**Soften** and can be **reshaped** again and again; harden when cooled	**Set hard** permanently the first time they are heated and shaped; they **do not soften** when heated again – they char (burn)
Structure	Long separate chains, weak forces between them	Chains joined by strong **cross-links**
Recycling	**Can be recycled** (melted and remoulded)	**Can't** be recycled by melting
Properties	Often flexible, softer; can be shaped with heat in the classroom	Hard, rigid, brittle, heat-resistant, good insulators
Examples	Acrylic (Perspex) – signs, display boxes; PET – cold-drink bottles; polyethene (HDPE/LDPE) – milk bottles, plastic bags; PVC – pipes, raincoats; polypropylene – crates, chairs; polystyrene – yoghurt tubs, packaging foam	Melamine – plates, kitchen counter tops; Bakelite / urea-formaldehyde – electrical plugs, switches, pot handles; epoxy resin – glue (e.g. Pratley), fibreglass boats

Manufacturing (processing) plastic products

Vacuum forming

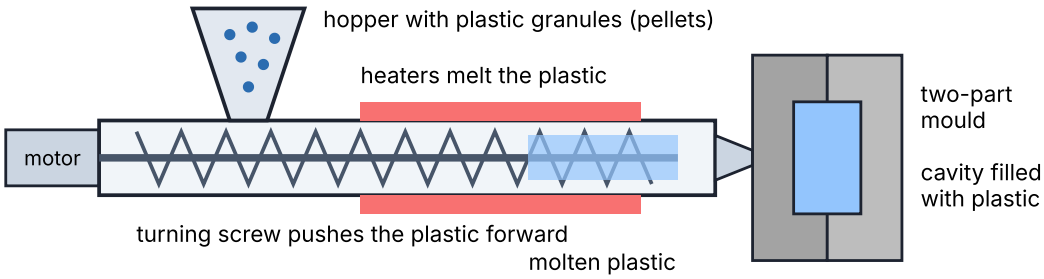


Vacuum forming

1. A thin **thermoplastic sheet** (often HIPS – high-impact polystyrene) is clamped in a frame above the mould.
2. The **heater** softens the sheet until it is floppy.
3. The **platen** with the **mould** on it is raised into the soft sheet.
4. The **vacuum pump** sucks the air out from underneath, so air pressure pushes the soft sheet **tightly over the mould**.
5. The plastic cools and hardens; the mould is removed and the edges trimmed.

The mould needs **sloping sides** (draft angle) and rounded corners so it can be removed. **Used for:** chocolate box inserts, yoghurt pots, blister packaging, masks.

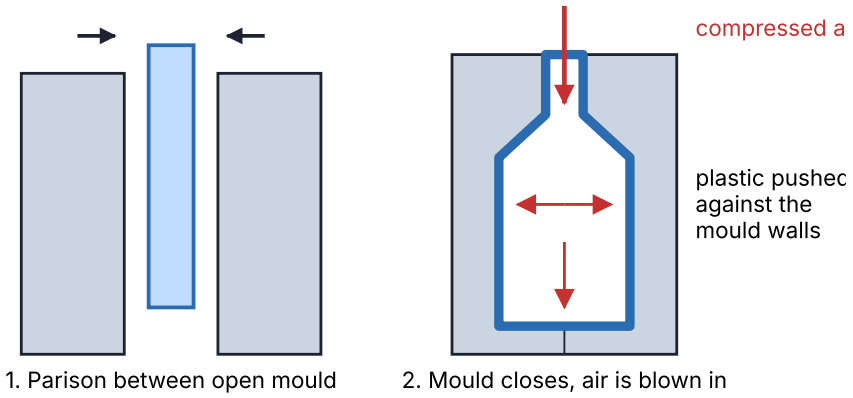
Injection moulding



Injection moulding

Thermoplastic **granules** are fed from the **hopper** into a heated barrel. A **screw** melts and mixes the plastic and then **forces (injects) it under high pressure** into a closed, cooled **metal mould**. The plastic cools and sets in the shape of the cavity, the mould opens and the product is pushed out. It is fast and very accurate – ideal for **mass production**. **Used for:** bottle caps, Lego blocks, buckets, chairs, phone covers, combs.

Blow moulding



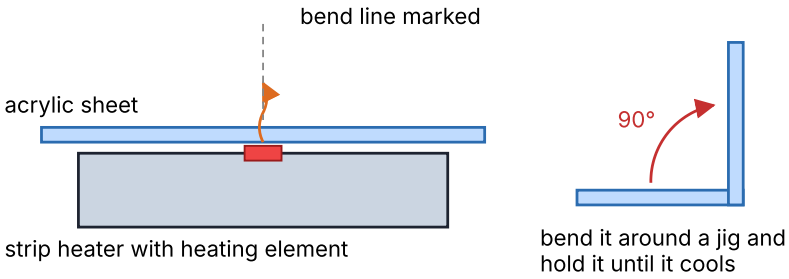
Blow moulding: 1. a hot tube of plastic (parison) hangs between the open mould halves; 2. the mould closes and air inflates it like a balloon

A soft, hot **tube** of thermoplastic (the **parison**) is placed between two mould halves. The mould closes and seals one end, then **compressed air is blown in** so the plastic expands like a balloon against the cold mould walls, cools and takes its shape. **Used for hollow products:** cold-drink (PET) bottles, milk bottles, shampoo bottles, plastic drums.

Extrusion

Like toothpaste squeezed from a tube: melted plastic is pushed by a screw through a shaped opening (**die**) to make long products with the **same cross-section** all along – **pipes, hoses, window frames, cable insulation, rods**. The product is cooled with water and cut to length.

Line bending (in the workshop)



Line bending a thermoplastic sheet

A **strip heater** heats a thermoplastic sheet (e.g. **acrylic**) along a **narrow line** only. When the plastic along the line is soft, it is bent to the required angle – often over a wooden **jig** or former so every bend is the same – and held until it cools and sets. Turn the sheet over every so often to heat both sides evenly. **Used for:** stands, phone holders, display signs, brackets.

Plastics and the environment

Problem	Explanation
Non-biodegradable	Most plastics are **not broken down by micro-organisms**; they can last **hundreds of years** in landfills and the environment.
Pollution of land and sea	Litter blocks drains; animals (turtles, seabirds) get entangled or eat plastic and die. Plastic breaks into tiny pieces (**microplastics**) that enter food chains.
Non-renewable resource	Made from crude oil, which will run out; making plastic uses energy and releases greenhouse gases.
Burning	Releases **toxic gases** (e.g. burning PVC) that are dangerous to breathe.

Solutions – the 3 Rs: Reduce (use less plastic, carry a reusable shopping bag), **Reuse** (use containers again), **Recycle** (sort plastics so they can be melted and made into new products). Also: **biodegradable plastics** (made from plants such as maize starch) that break down more quickly.

Recycling codes

Plastic products are marked with a **number inside a triangle of arrows** to show which type of plastic it is, so it can be sorted for recycling.



Plastic recycling (identification) codes

Key terms

Term	Meaning
Design brief	A short statement of what will be designed and made, for whom and why.
Specifications	The requirements that the product must meet.
Constraints	The limits within which the design must be done (e.g. cost, time, materials).
Evaluate	To test and judge a product against its specifications.
Isometric drawing	A 3D drawing where vertical lines stay vertical and horizontal lines are drawn at 30°.
Orthographic drawing	A set of 2D views (front, top, side) of an object, drawn with exact sizes.
First-angle projection	The orthographic layout in which the top view is below the front view and the left view is to the right of it.
Hidden line	A dashed line that shows an edge you can't see from that side.
Circuit	A closed path through which electric current flows.
Resistance	The opposition to the flow of current, measured in ohms (Ω).
Ohm's law	$V = I \times R$: current is directly proportional to voltage at constant temperature.
LDR	A resistor whose resistance decreases as light intensity increases.
Thermistor	A resistor whose resistance changes with temperature.
Diode / LED	A component that lets current flow in one direction only; an LED also gives off light.
Capacitor	A component that stores electric charge.
Transistor	An electronic switch: a small base current switches on a larger collector–emitter current.
Logic gate	A control device that gives an output (1 or 0) depending on its inputs (AND, OR, NOT, NAND, NOR).
Truth table	A table that shows the output of a logic gate for every combination of inputs.
Polymer	A long-chain molecule made of many repeating units (monomers).
Thermoplastic	A plastic that softens when heated and can be reshaped and recycled.
Thermosetting plastic	A plastic that sets permanently hard when first heated and can't be reshaped.
Biodegradable	Can be broken down naturally by micro-organisms.